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Topological exciton insulator

Zhongqi Xiu

2026.09.10

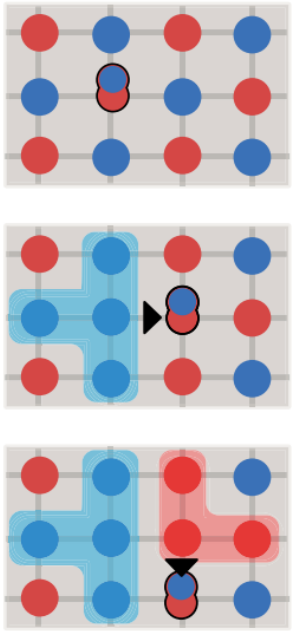
flow

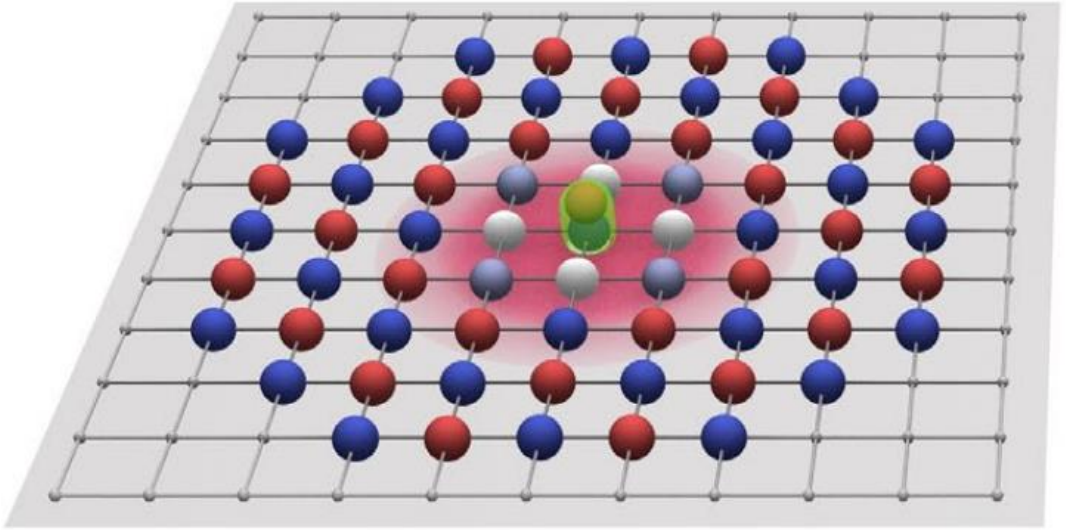
1.Magnetic polaron picture()
2.CrSBr property: AFM FM, axis, crystal and lattice structure
3.Interlayer exciton energy can be tuned with B/E
4.How exciton affect magnetic (model in papers)
5.For magnetic polaron, how it exsist in this system?
6.How can we identify it?

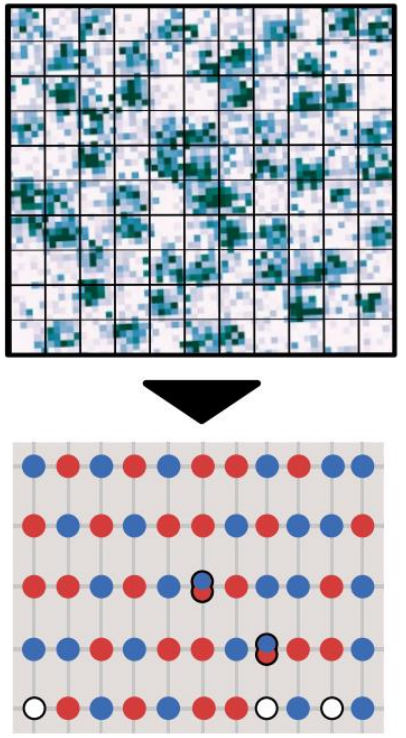
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arXiv:2509.13159

Magnetic Polaron







Full reconstruction

Impurity: dopants such as holes move around
Bath: magnetic order/ spin system generated by ½ fermions close to half filling in lattice

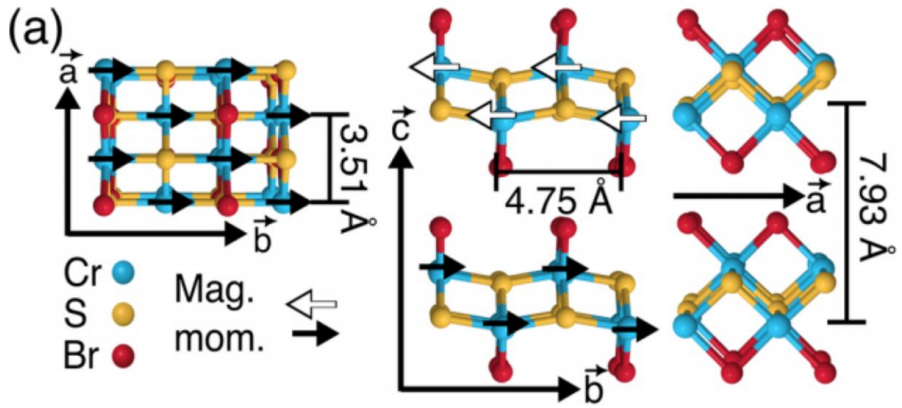
Magnetic polaron is defined as the coupled state of the moving dopants and affected environment

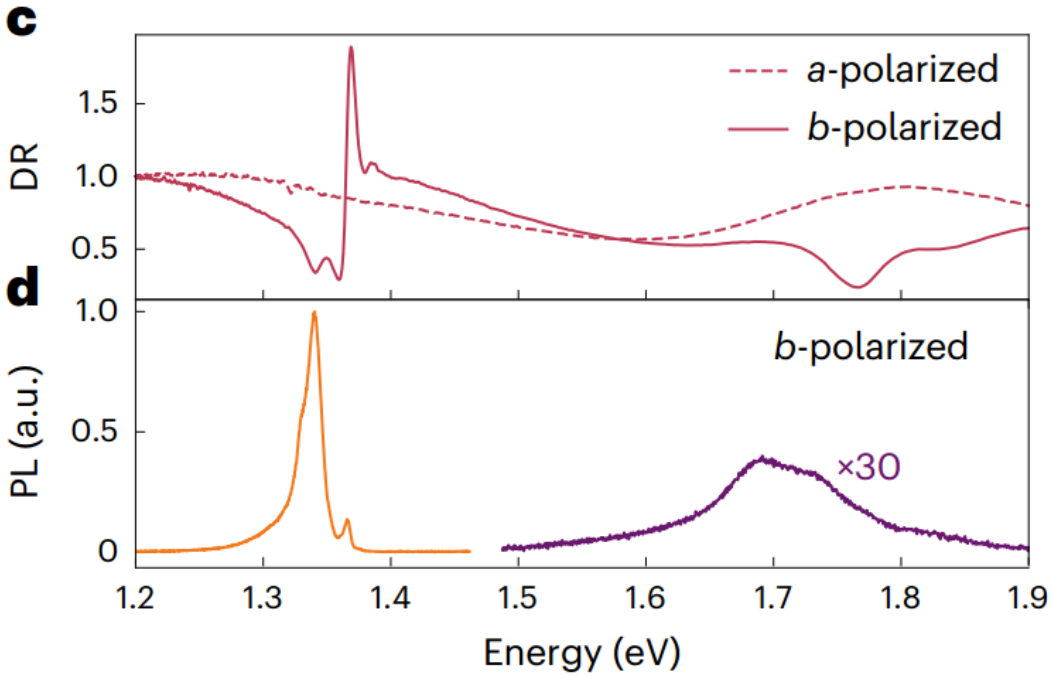
Hole in AFM state?

Nature 572, 358–362 (2019)

CrSBr crystal structure

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- 每层内部是 FM，相邻层之间是 AFM。
- \(b\) 是 easy axis，\(a\) 是 intermediate axis，\(c\) 是 out-of-plane hard axis。
- 沿不同方向施加磁场，可以产生不同的翻转或连续 canting。
- 层间交换和磁各向异性共同决定自旋偏转的代价。

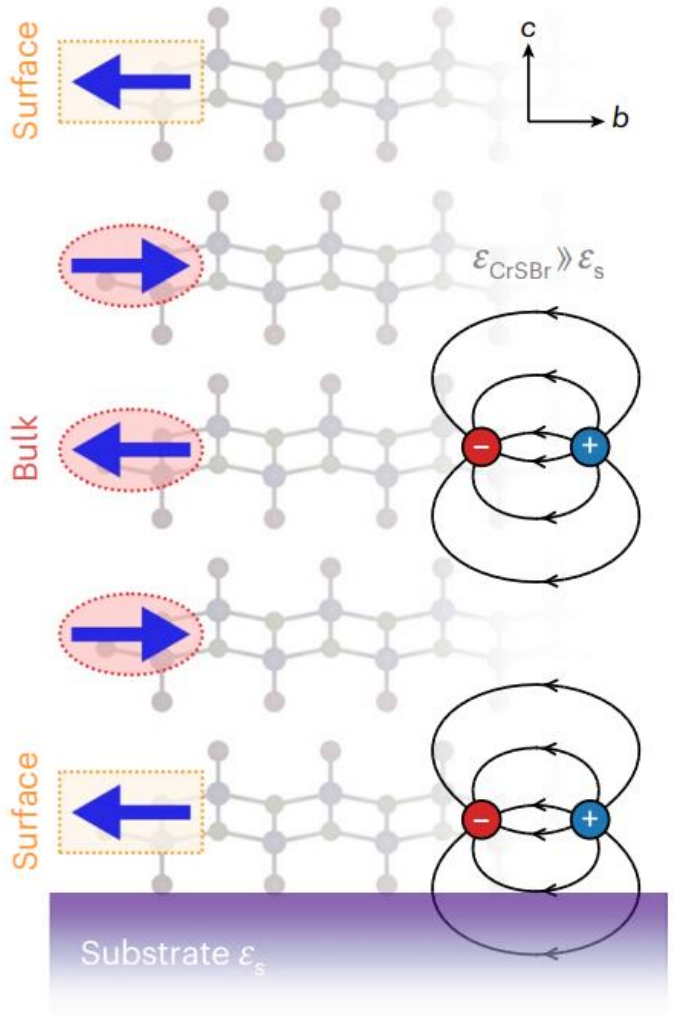
这一页最重要的句子是：

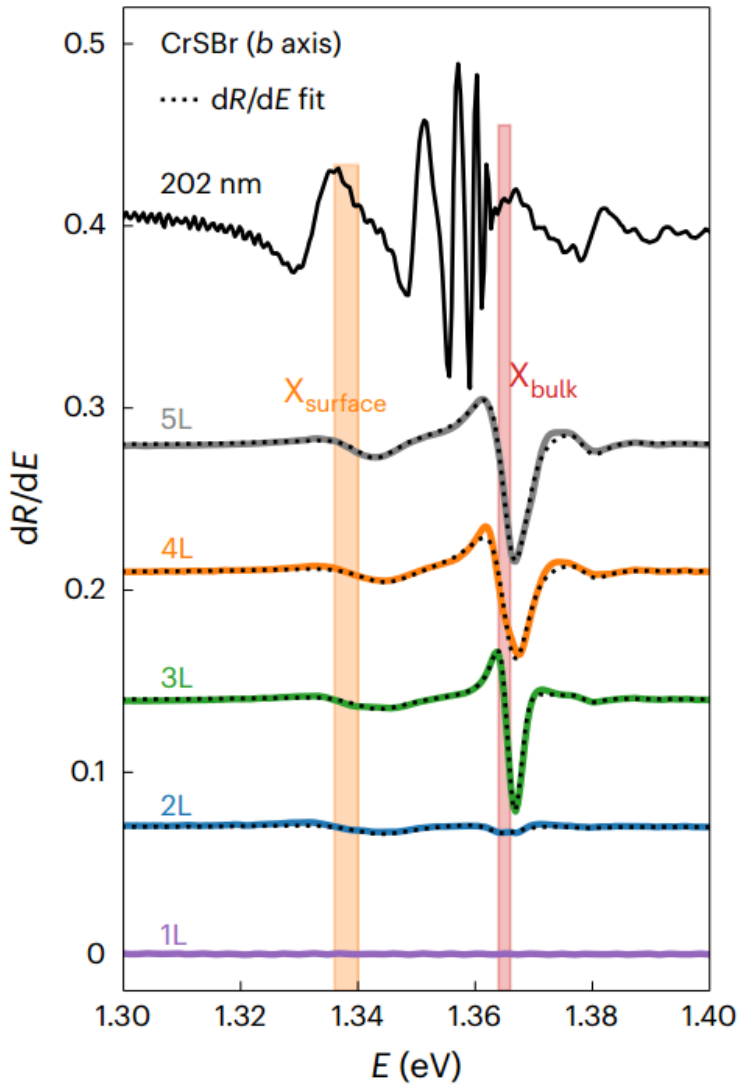
CrSBr 给我们一个可以连续调节的层间自旋夹角；激子可以感受到这个角度，自旋本身又有抵抗偏转的刚度。

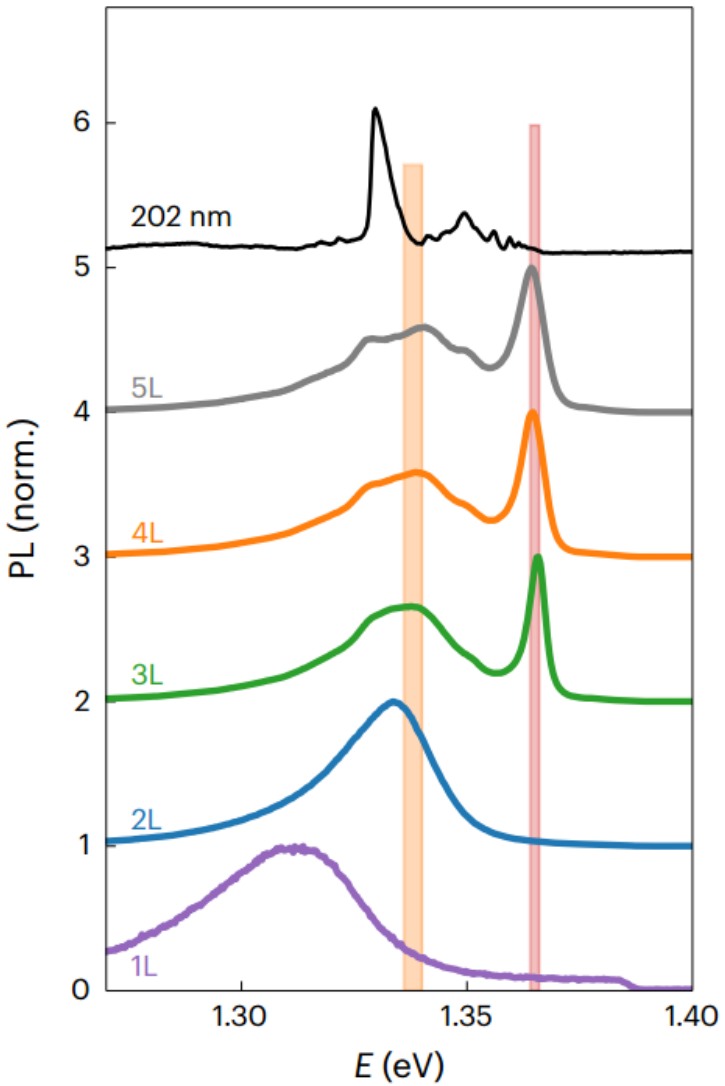
Nat. Mater. 24, 1027–1033 (2025)

Surface and bulk excitons

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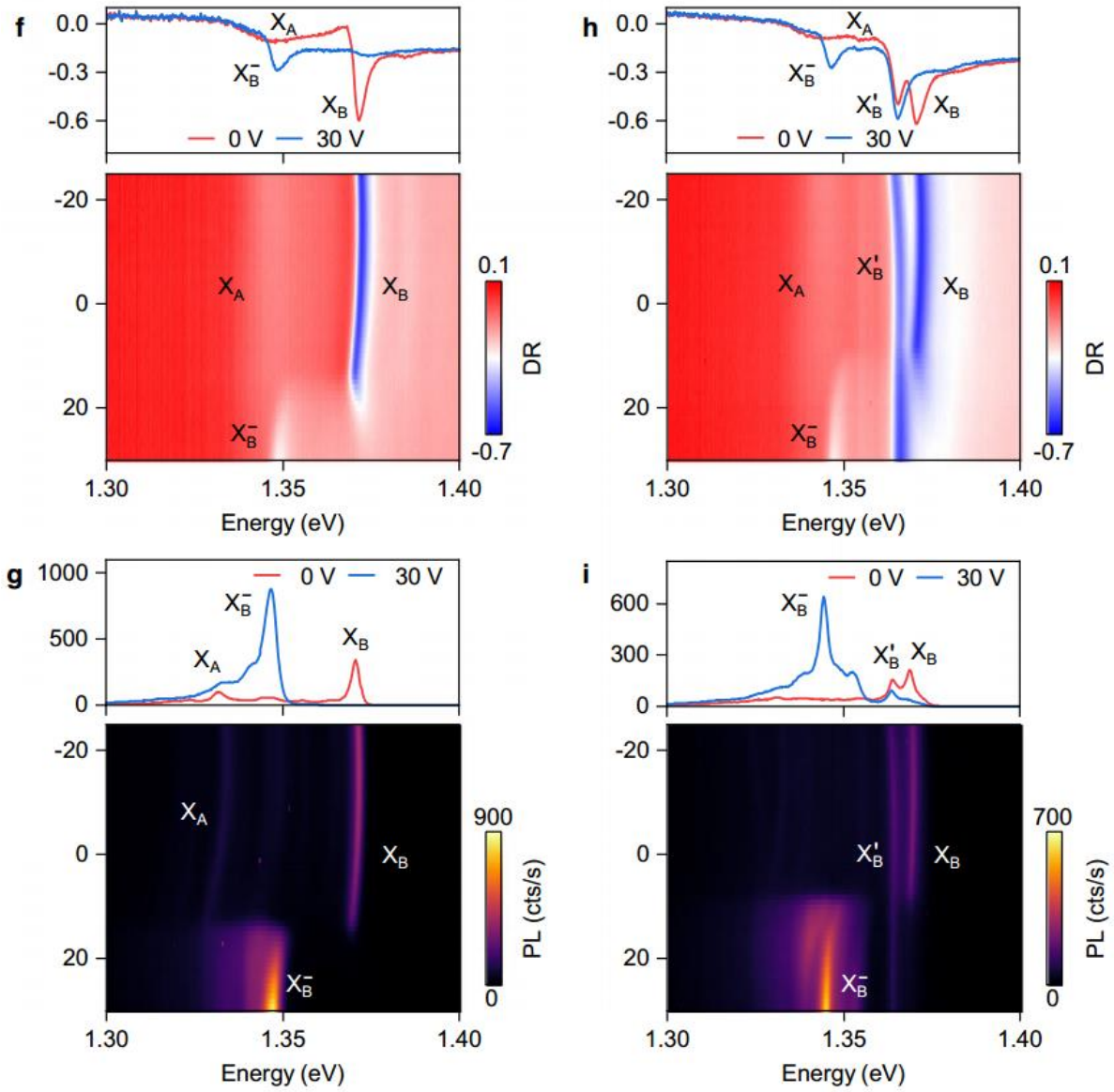






Nat. Mater. 24, 391–398 (2025).

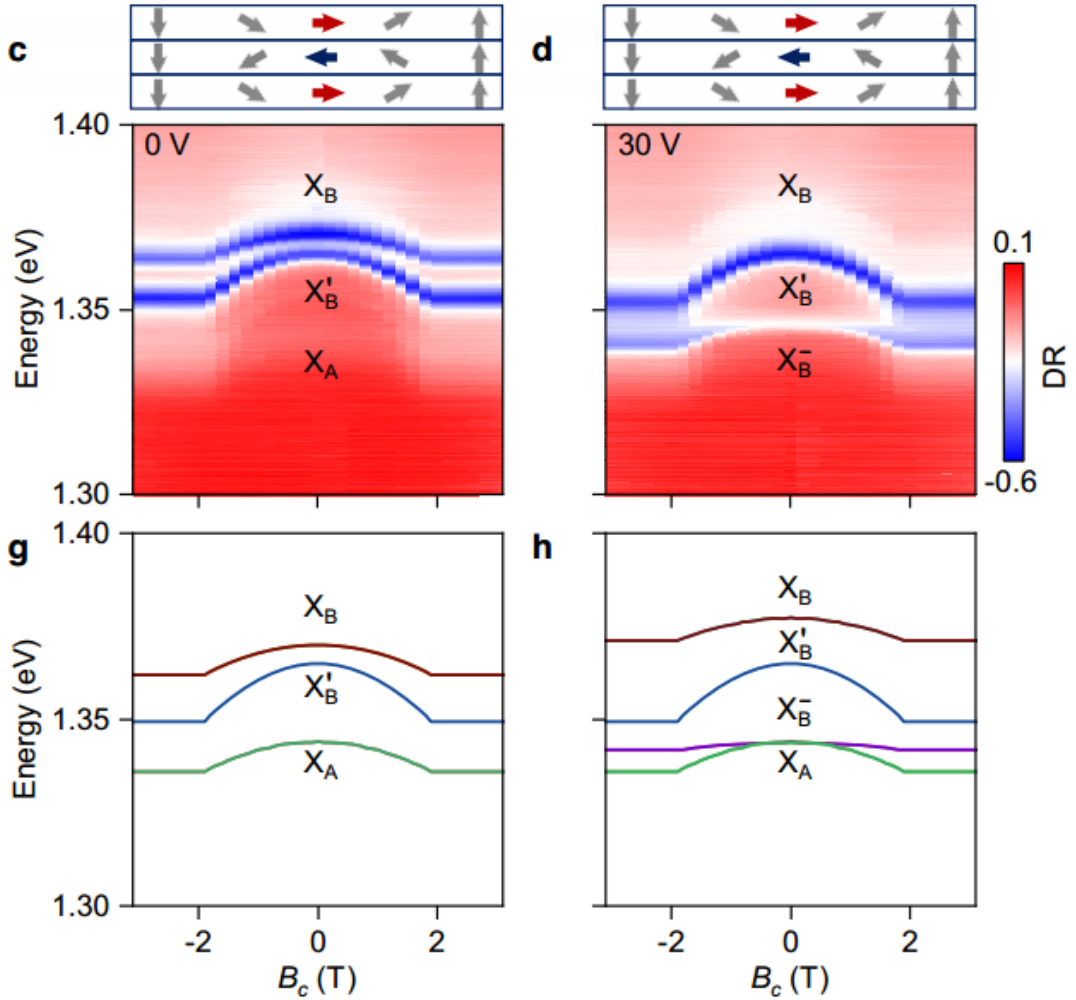
Exciton related to doping



通过fermipolaron讲明白这些exciton的energy

Nat Commun 15, 4735 (2024)

Exciton related to B field



$$t_{\text{eff}} \propto \cos\frac{\theta}{2}, \qquad E_X(\theta) = E_{X,0} - \Delta\cos^2\frac{\theta}{2}.$$

激子能量随磁排列改变；不同激子对磁排列的敏感度不同。
如果自旋转动能够降低激子的能量，那么加入激子以后，它会不会推动自旋往能量更低的方向转动

Nat Commun 15, 4735 (2024).

Device structure

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How the magnetic polaron form or exsist in this system.
Focus on how to identify this localized magnetic polaron species in this system?
How to measure the response of this

Device structure

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没有激子时，自旋选择磁性能量最低的位置。加入激子后，总能量多了一个依赖角度的部分。激子通过更有利的层间杂化获得能量，因此推动自旋偏转；磁背景则通过交换和各向异性抵抗这种偏转。

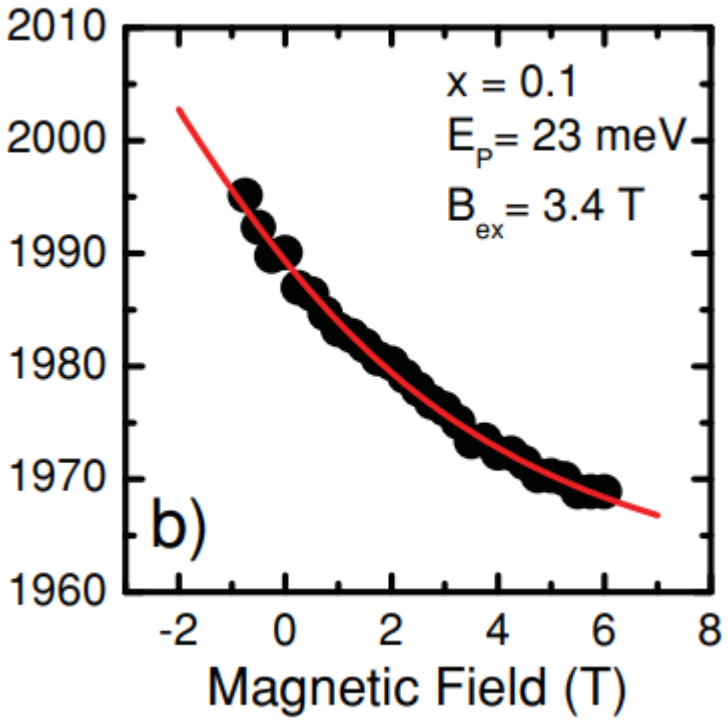
Photoluminescence and reflectivity measurements

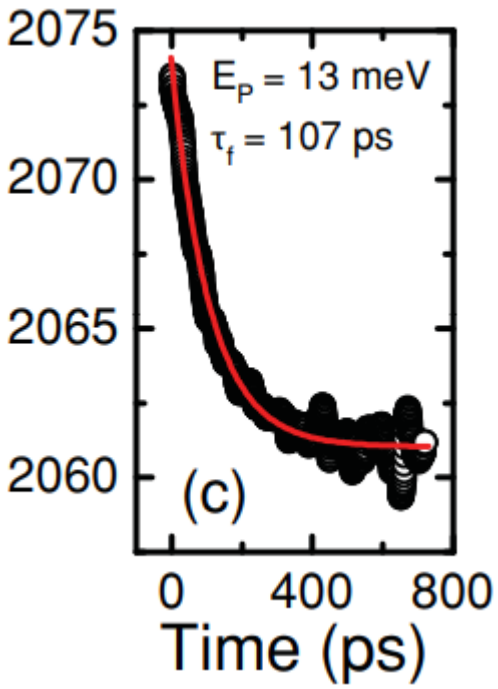
PL and reflectivity measurements were conducted on the CrSBr sample using a Montana cryostat at a base temperature of 4 K. For the PL measurements, the sample was excited using a continuous-wave (CW) diode laser at a wavelength of 532 nm. Reflectivity measurements utilized a broadband CW white light source. A ×100 objective (Mitutoyo M Plan Apo) was used to focus the incident light onto the sample and to collect both the reflected light and the PL signal. This collected signal was subsequently analysed using a spectrometer (Model SpectraPro HRS-500, Princeton Instruments) equipped with a grating of 300 lines per millimetre.

arXiv:2509.13159

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Phys. Rev. B 83, 081306(R)

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THANKS

Speaker: Zhongqi Xiu

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